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smolyax: a high-performance implementation of the Smolyak interpolation operator in JAX

Josephine Westermann [1¶] **and Joshua Chen** [2]

1 Heidelberg University, Germany **2** The University of Texas at Austin, USA ¶ Corresponding author

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Summary

The `smolyax` library provides interpolation capabilities for arbitrary multivariate and vector-valued functions $f : \mathbb{R}^{d_1} \to \mathbb{R}^{d_2}$ for any $d_1, d_2 \in \mathbb{N}$.

It implements the Smolyak interpolation operator, which is known to overcome the curse-of-dimensionality plaguing naive multivariate interpolation (Barthelmann et al., 2000) and uses the barycentric interpolation formula (Berrut & Trefethen, 2004) for numerical stability. The implementation is based on JAX (Bradbury et al., 2018), a free and open-source Python library for high-performance computing that integrates with the Python and NumPy numerical computing ecosystem. Thanks to JAX's device management, `smolyax` runs natively on both CPU and GPU. While implementing Smolyak interpolation in JAX is challenging due to the highly irregular data structures involved, `smolyax` overcomes this by employing a tailored batching and padding strategy that enables efficient vectorization.

`smolyax` supports interpolation on bounded or unbounded domains via Leja (M. A. Chkifa, 2013) or Gauss-Hermite (Abramowitz & Stegun, 1964) node sequences, respectively. It provides efficient Numba-accelerated routines to generate isotropic or anisotropic total degree multi-index sets (Adcock et al., 2022, sec. 2.3.2), which are the key ingredient to generate the high-dimensional sparse grids (Bungartz & Griebel, 2004) of interpolation nodes required by the Smolyak interpolation operator. Additional types of node sequences or multi-index sets can be incorporated easily by implementing a minimalistic interface. The `smolyax` interpolant provides further functionality to evaluate its gradient as well as compute its integral.

Statement of Need

Polynomial expansion is a well-studied and powerful tool in applied mathematics, with important applications in surrogate modeling, uncertainty quantification and inverse problems, see e.g. (Adcock et al., 2022; A. Chkifa et al., 2015; Dũng et al., 2023; Herrmann et al., 2024; Westermann et al., 2025; Zech, 2018) and references therein.

Smolyak interpolation offers a practical way to construct polynomial approximations with known error bounds for a wide range of function classes, see e.g. (Barthelmann et al., 2000; A. Chkifa et al., 2015).

While several libraries provide high-dimensional interpolation functionality, none, to our knowledge, provides a hardware-agnostic, high performance implementation. `smolyax` addresses this gap by providing an efficient solution within the popular JAX ecosystem.

A vectorizable implementation of the Smolyak operator

Recall that given a domain $D \subset \mathbb{R}$ and set of $\nu + 1 \in \mathbb{N}$ pairwise distinct interpolation points $(\xi_i^\nu)_{i=0}^\nu \subset D$, the univariate polynomial interpolation operator I^ν maps a function $f : D \to \mathbb{R}$ onto the unique polynomial $I^\nu[f] \in \mathbb{P}_\nu := \operatorname{span}\{x^i \,:\, i = 0, \dots, \nu\}$ of maximal degree ν such that $f(\xi_i^\nu) = I^\nu[f](\xi_i^\nu)$ for all $i \in \{0, 1, \dots, \nu\}$.

Tensorized interpolation generalizes univariate interpolation to multivariate functions defined on a tensor-product domain $D = \otimes_{j=1}^d D_1$ with $D_1 \subset \mathbb{R}$ and $d \in \mathbb{N}$ by defining $I^{\boldsymbol{\nu}} := \otimes_{j=1}^d I^{\nu_j}$, where $\boldsymbol{\nu} \in \mathbb{N}_0^d$ is a multi-index characterizing the maximal polynomial degree in each dimension. Since $I^{\boldsymbol{\nu}}[f] \in \mathbb{P}_{\boldsymbol{\nu}} := \operatorname{span}\{\boldsymbol{x}^{\boldsymbol{\mu}} \,:\, \boldsymbol{\mu} \leq \boldsymbol{\nu}\}$, this approach suffers from the curse of dimensionality as d increases.

Smolyak interpolation (Barthelmann et al., 2000; Smolyak, 1963) overcomes this issue by introducing polynomial ansatz spaces $\mathbb{P}_\Lambda := \operatorname{span}\{\boldsymbol{x}^{\boldsymbol{\mu}} \,:\, \boldsymbol{\mu} \in \Lambda\}$ parametrized by downward closed multi-index sets $\Lambda \subset \mathbb{N}_0^d$. The resulting interpolation operator is a linear combination of tensorized interpolation operators:

$$I^\Lambda := \sum_{\boldsymbol{\nu} \in \Lambda} \zeta_{\Lambda,\boldsymbol{\nu}} I^{\boldsymbol{\nu}}, \qquad \zeta_{\Lambda,\boldsymbol{\nu}} := \sum_{\boldsymbol{e} \in \{0,1\}^d : \boldsymbol{\nu} + \boldsymbol{e} \in \Lambda} (-1)^{|\boldsymbol{e}|}. \tag{1}$$

Implementing this operator in a vectorized form suitable for high-performance computing is nontrivial, as vectorization requires inputs to conform to a uniform structure. However, each tensorized interpolant in (1) involves reducing a higher-order tensor of unique shape $\boldsymbol{\nu}$ via multiplication with one vector per dimension. A naive strategy would be to zero-pad all tensors in (1) to the smallest common shape $(\max_{\boldsymbol{\nu} \in \Lambda}(\nu_j))_{j=1}^d$. This, however, reintroduces the curse of dimensionality, as memory requirements grow exponentially with d.

`smolyax` strikes a balance between handling small, uniquely shaped tensors and large, identically shaped ones. The key idea is to group tensors by their number of active dimensions and prepare them for vectorized processing within each group. In particular, this involves

1. Dropping indices ("*squeezing*") of non-active dimensions, i.e., j with $\nu_j = 0$.
2. Permuting the remaining active dimensions in descending order, and
3. Zero-padding all tensors with the same number of active dimensions to the smallest common shape.

This reorganizes the tensors into a small number of large, structured data blocks that can be processed in a fast, vectorized way. Asymptotically in both the dimension and the size of the polynomial space, this approach does not consume more memory than the original bare tensors, up to a small constant.

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